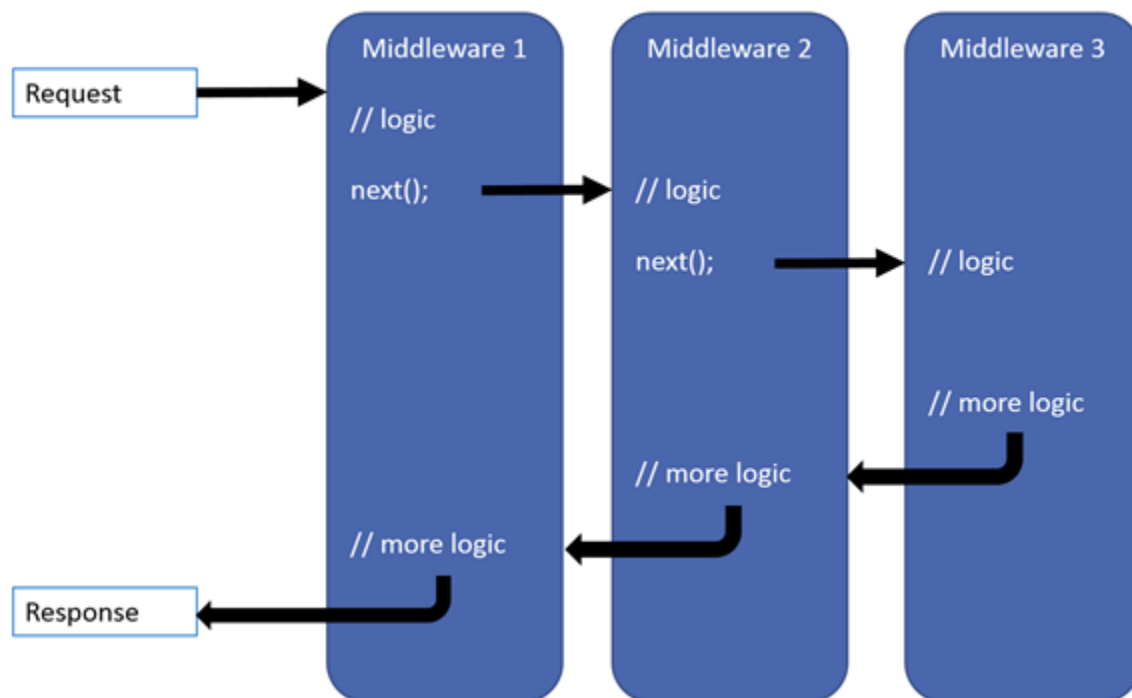


✓ What is Middleware in ASP.NET Core?

Middleware is a component (a small piece of code) in the ASP.NET Core request pipeline that:

- Receives an HTTP request
- Performs some processing
- Either **handles the request** OR **passes it to the next middleware**
- Finally processes the response on the way back

Think of middleware as **filters** arranged in a pipeline:



📌 Key Points About Middleware

1. Middleware runs **in the order** they are added in Program.cs.
2. Each middleware has:
 - ✓ Request logic
 - ✓ Optionally response logic

3. You can:

- Use built-in middleware
- Write custom middleware
- Add terminal middleware (which ends the pipeline)

How Middleware Pipeline Works

Example pipeline:

```
app.UseMiddleware<A>();
```

```
app.UseMiddleware<B>();
```

```
app.UseMiddleware<C>();
```

Execution Flow:

→ Request → A → B → C → (Endpoint)

← Response ← A ← B ← C

Each middleware can:

- Do something **before** passing to next component
- Do something **after** next component completes

★ Types of Middleware

✓ Use

Runs code **before & after** next middleware.

```
app.Use(async (context, next) =>
```

```
{
```

```
    // before
```

```
    await next(); // pass to next
```

```
    // after
```

```
});
```

✓ Run

Terminal middleware — **does not call next()**

```
app.Run(async context =>
{
    await context.Response.WriteAsync("Pipeline ended here!");
});
```

✓ Map / MapWhen

Conditional branching in pipeline.

```
app.Map("/admin", adminApp =>
{
    adminApp.Run(async context =>
        await context.Response.WriteAsync("Admin area"));
});
```

🌱 Built-in Middleware Examples

Middleware	Purpose
UseRouting	Maps request URL to endpoints
UseAuthentication	Validates user identity
UseAuthorization	Checks permissions
UseStaticFiles	Serves CSS, JS, images
UseCors	Configures cross-origin access
UseExceptionHandler	Global error handling

🧠 Why Middleware Is Important?

You use middleware for global concerns such as:

- ✓ Logging
- ✓ Authentication / Authorization
- ✓ Exception handling
- ✓ Response compression
- ✓ Custom headers
- ✓ IP blocking / rate limiting
- ✓ Static file handling
- ✓ Routing

Middleware helps you build pipelines where each component does one task **cleanly and independently**, following **Single Responsibility Principle**.

Real-Time Example Pipelines

E-commerce Website Request

1. UseHttpsRedirection
 2. UseRouting
 3. UseAuthentication → validate user login
 4. UseAuthorization → check user roles
 5. UseCors
 6. Custom middleware: log product requests
 7. MVC Controller → /products/5
-

Final Summary

Middleware = Small components that process HTTP requests in order.

- ✓ Order matters
- ✓ Each middleware can:

- Inspect request
- Modify response
- Call next
- Stop pipeline

✓ Can be built-in or custom

Middleware makes ASP.NET Core **fast, modular, and powerful**.